Title of the project work

Name of the student[1], Name of the student[2], Name of the student[3]

Email id of the 1st student, Email id of the 2nd student, Email id of the 3rd student

1. Introduction

Write one page about your project title.

2. Literature Survey

The comparative analysis of the recent studies works on sentiment analysis of drug reviews is provided below.

Sahar Eskandarisani and Mohammad Reza Keyvanpour [1] adopted a stacking ensemble approach to sentiment analysis of drug reviews by Random Forest, Logistic Regression, and MLP with XGBoost as a meta-model. The TF IDF feature among the TF IDF, Bag of Words and CBOW features, gave the best results, with an accuracy of 89%. The model outperformed the CatBoost, LightGBM, and single learners and provides support for healthcare decision-making based on their insights.

Anuj Kumar [2] used LightGBM, Random Forest and Decision Tree classifiers to classify the sentiments as positive and negative by analyzing 16,297 pharmaceutical reviews. Random Forest resulted in the highest accuracy (90%) which helped understand experiences of patients taking drugs and making improvements in drugs.

V. Lakshman Narayana et al. [3] proposed a medication recommending system based on sentiment analysis and machine learning. A Priority Weight Allocation Model with User Ratings and Gradient Boosting on TF-IDF achieved 98% accuracy which resulted in accurate and side-effect aware drug recommendations hence resulting in improved patient care and satisfaction.

Table 1: Literature Survey

S.No.	Title & Year	Dataset Used	Algorithm Used	Performance Measure	Future Work
1.	Sentiment Analysis on Drug Reviews Using Ensemble Learning Method (2024)	**Link of dataset & Size of dataset to be written**	**Name of algorithm to be written**	**Corresponding algorithm Performance Measure like accuracy, precision, recall, F1 Score or F-Measure to be written**	**Copy paste the future work present in the paper and should be highlighted with yellow colour in the pdf file**
2.	Machine Learning Based Sentiment Analysis of Pharmaceutical Reviews (2024)				
3.	Drug Recommendation System and Side Effects Based on Sentiment Analysis of Drug Reviews Using Machine Learning (2024)				

3. References

1. Sahar Eskandarisani and Mohammad Reza Keyvanpour, “Sentiment Analysis on Drug Reviews Using Ensemble Learning Method,” in *Proc. 2024 20th Int. Symp. Artificial Intelligence and Signal Processing (AISP)*, Mazandaran University of Science and Technology, Babol, Iran, Feb. 2024, pp. 1–6, doi: 10.1109/AISP61396.2024.10475227.
2. Anuj Kumar, *“Machine Learning Based Sentiment Analysis of Pharmaceutical Reviews,”* in Proc. 2024 5th Int. Conf. Electronics and Sustainable Communication Systems (ICESC), Coimbatore, India, Jul. 2024, pp. 1850–1855, doi: 10.1109/ICESC60852.2024.10689892
3. V. Lakshman Narayana, K. Sai Nikitha, SK Fathima Yaseen, P. Madhu Latha, and T. G. Sahithi, *“Drug Recommendation System and Side Effects Based on Sentiment Analysis of Drug Reviews Using Machine Learning,”* in Proc. 2025 4th Int. Conf. Smart Technologies, Communication and Robotics (STCR), Sathyamangalam, India, May 2025, pp. 1–6, doi: 10.1109/STCR62650.2025.11019585
4. Muhammad Swaileh A.Alzaidi, Alya Alshammari, Manar Almanea, Haseen A. Al-Khawaja, Hanan Al Sultan, Shoayee Alotaibi, and Wafa Almukadi, *“A Text-Inception-Based Natural Language Processing Model for Sentiment Analysis of Drug Experiences,”* ACM Trans. Asian Low-Resour. Lang. Inf. Process., Aug. 2024, doi: 10.1145/3678470
5. Sena Al-Hadhrami, Tamas Vinko, Tawkif Al-Hadhrami, Faisal Saeed, and Sultan Noman Qasem, “Deep learning-based method for sentiment analysis for patients’ drug reviews,” *PeerJ Comput. Sci.*, vol. 10, Apr. 2024, Art. e1976, doi: 10.7717/peerj-cs.1976

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In that folder, save all pdf files as year_title of the paper